# SHEAVES ON THE ETALE SITE OF A VARIETY OVER A FINITE FIELD ADMIT CANONICAL FROBENIUS DESCENT ISOMORPHISMS

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*Allowed standard prerequisites (editorial).*

- The proof of Theorem theorem-baffling exhibits, for every etale varphi : U -> X, a functorial isomorphism U \cong U \times_{varphi, X, F_X} X over X (author theorem environment and its proof, source lines 321-331 and 344-359). invoked at proof lines 392-394, with pi_X = F_X^f the geometric frobenius. Supplies the functorial isomorphism gamma_U the sheaf restriction is taken along.

*Source reference.* Original theorem 03SN: sources/stacks-chow/tags.txt, source lines 4974–4974. Invoked at proof lines 392-394 ('it follows from the proof of Theorem baffling') for the functorial isomorphism gamma_U; load-bearing inside the counted span.

**Lemma 0.2.** *Let $\mathcal{F}$ be a sheaf on $X_{étale}$. Then there are canonical isomorphisms $\pi_X^{-1}\mathcal{F} \cong \mathcal{F}$ and $\mathcal{F} \cong \pi_{X*}\mathcal{F}$.*

*Proof.* Let $\varphi : U \to X$ be étale. Recall that $\pi_{X*}\mathcal{F}(U) = \mathcal{F}(U \times_{\varphi,X,\pi_X} X)$. Since $\pi_X = F_X^f$, it follows from the proof of Theorem 03SN that there is a functorial isomorphism

$$\begin{array}{ccc} U & \xrightarrow{\gamma_U} & U \times_{\varphi,X,\pi_X} X \\ & {\scriptstyle\varphi}\searrow \quad \swarrow{\scriptstyle \mathrm{pr}_2} & \\ & X & \end{array}$$

where $\gamma_U = (\varphi, F_U^f)$. Now we define an isomorphism

$$\mathcal{F}(U) \longrightarrow \pi_{X*}\mathcal{F}(U) = \mathcal{F}(U \times_{\varphi,X,\pi_X} X)$$

by taking the restriction map of $\mathcal{F}$ along $\gamma_U^{-1}$. The other isomorphism is analogous. □

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*Date*: stacks-project:trace.tex:local-snapshot-20261012; GNU FDL 1.2 or later.